Reproduction

Human Physiology - Chapter 20

Biology Teaching Resources

December 24, 2025

Outline

Outline

Sex Determination

- **Sexual reproduction** (有性生殖) involves fusion of **gametes** (配子) - sperm and ova
- **Germ cells** (生殖细胞) formed in **gonads** (性腺) through **meiosis** (减数分裂)
- Chromosome number reduced from 46 to 23 in gametes
- **Fertilization** (受精) restores diploid number (46) in **zygote** (受精卵)

Sex Determination

- The human life cycle alternates between haploid and diploid states
- Meiosis produces haploid gametes (23 chromosomes)
- Fertilization produces diploid zygote (46 chromosomes)
- This cycle maintains genetic stability across generations

Sex Determination

- **Chromosomal sex** (染色体性别) determined at conception
- Females: XX chromosomes
- Males: XY chromosomes
- Y chromosome contains **SRY gene** (性别决定区基因) encoding **TDF** (testis-determining factor, 睾丸决定因子)

Sex Determination

- The Y chromosome is much smaller than X chromosome
- Despite small size, Y chromosome carries critical genes for male development
- SRY gene is the master regulator of sex determination

Sex Determination

- **X-inactivation** (X 染色体失活) occurs in female cells
- One X chromosome becomes inactive forming **Barr body** (巴氏小体)
- This achieves **dosage compensation** (剂量补偿) between males and females
- Random inactivation leads to mosaic pattern in females

Sex Determination

- Early embryos have **indifferent gonads** (未分化性腺)
- Gonads can develop into either testes or ovaries
- Presence of TDF directs testis development (week 7)
- Without TDF, ovaries develop by default (week 9-10)

Development of Accessory Sex Organs

- Two duct systems present in early embryo:
  - **Wolffian ducts** (沃尔夫管) → male structures
  - **Müllerian ducts** (苗勒管) → female structures
- Hormones from gonads determine which system develops

Development of Accessory Sex Organs

- In males: testosterone stimulates Wolffian duct development
- **MIF** (müllerian inhibition factor, 苗勒管抑制因子) causes Müllerian duct regression
- In females: without testosterone and MIF, Müllerian ducts develop
- Wolffian ducts regress without testosterone

Development of Accessory Sex Organs

- External genitalia develop from common primordia
- Genital tubercle, urogenital folds, labioscrotal swellings
- Male development (with testosterone): tubercle → penis; folds fuse; swellings → scrotum
- Female development (without testosterone): tubercle → clitoris; folds → labia minora; swellings → labia majora

Development of Accessory Sex Organs

- Developmental timetable shows critical periods for sexual differentiation
- Sex determination → gonad development → duct system development → external genitalia
- Most differentiation complete by week 12 of development

Development of Accessory Sex Organs

- **DHT (dihydrotestosterone)** (双氢睾酮) is more potent than testosterone
- Enzyme **5α-reductase** (5α-还原酶) converts testosterone to DHT
- DHT essential for male external genitalia and prostate development

Disorders of Sexual Development

- **Androgen insensitivity syndrome (AIS)** (雄激素不敏感综合征): XY individuals with non-functional receptors develop female external genitalia
- **5α-reductase deficiency** (5α-还原酶缺乏症): XY individuals cannot make DHT, resulting in ambiguous genitalia
- **Congenital adrenal hyperplasia (CAH)** (先天性肾上腺增生): XX individuals exposed to excess androgens
- These reveal critical roles of hormones and receptors in development

Outline

Steroid Hormone Biosynthesis

- **Steroid hormones** (类固醇激素) derived from cholesterol
- Common biosynthetic pathway for all steroid hormones
- Testes and ovaries can produce both androgens and estrogens
- Differ in proportions produced

Hypothalamic-Pituitary-Gonadal Axis

- **HPG axis** (下丘脑-垂体-性腺轴) regulates reproduction
- **GnRH** (gonadotropin-releasing hormone, 促性腺激素释放激素) from hypothalamus
- Stimulates anterior pituitary to secrete FSH and LH
- **FSH** (follicle-stimulating hormone, 促卵泡激素)
- **LH** (luteinizing hormone, 促黄体激素)

Hypothalamic-Pituitary-Gonadal Axis

- **Negative feedback** (负反馈) maintains hormonal homeostasis
- Sex steroids inhibit GnRH and gonadotropin secretion
- This feedback loop regulates reproductive function
- Balance between stimulation and inhibition

Puberty

- **Puberty** (青春期) initiated by increased GnRH secretion
- Occurs in pulsatile fashion
- Hypothalamic sensitivity to negative feedback decreases
- Allows increased gonadotropin secretion
- Triggers gonadal maturation and sex steroid production

Puberty

- Girls' puberty typically begins age 10-12
- **Thelarche** (乳房发育): breast development
- **Pubarche** (阴毛发育): pubic and axillary hair growth
- **Menarche** (初潮): first menstrual period
- Changes in body composition and fat distribution

Puberty

- Boys' puberty typically begins age 12-14
- Testicular and penile growth
- Voice deepening due to laryngeal growth
- Facial and body hair growth
- Increased muscle mass and bone density
- Onset of spermatogenesis

Puberty

- Growth spurt timing differs between sexes
- Females: growth spurt at age 11-12
- Males: growth spurt at age 13-14
- Girls temporarily taller than boys in early adolescence
- Males eventually achieve greater adult height

Other Regulatory Factors

- **Pineal gland** (松果体) secretes **melatonin** (褪黑素)
- Melatonin may inhibit GnRH and delay puberty
- Photoperiod influences reproductive seasonality in many animals
- **Libido** (性欲) influenced by testosterone in both sexes

Outline

Testicular Structure

- **Testes** (睾丸) produce sperm and testosterone
- **Seminiferous tubules** (曲细精管): site of spermatogenesis
- **Interstitial (Leydig) cells** (间质细胞): secrete testosterone
- Spatial separation allows independent regulation

Hormonal Control

- FSH acts on **Sertoli cells** (支持细胞) to support spermatogenesis
- LH stimulates Leydig cells to secrete testosterone
- Testosterone provides negative feedback on LH
- **Inhibin** (抑制素) from Sertoli cells inhibits FSH

Hormonal Control

- Dual feedback system allows independent regulation
- Testosterone production can be regulated separately from spermatogenesis
- FSH and inhibin regulate sperm production
- LH and testosterone regulate androgen levels

Testosterone Metabolism

- Testosterone converted to more potent forms in target tissues
- **5α-reductase** (5α-还原酶) converts testosterone to DHT
- **Aromatase** (芳香化酶) converts testosterone to estradiol
- Different metabolites mediate different effects

Testosterone Metabolism

- Androgens have diverse physiological effects
- Development of reproductive organs during embryonic life
- Secondary sex characteristics at puberty
- Protein anabolism and muscle growth
- Bone growth and maintenance
- Behavioral and cognitive effects

Testosterone Metabolism

- Testosterone acts within testes via paracrine mechanisms
- Intratesticular concentration 100× higher than blood
- High local concentration essential for spermatogenesis
- Demonstrates importance of local hormone action

Spermatogenesis

- Spermatogenesis takes approximately 74 days
- **Spermatogonia** (精原细胞) are diploid stem cells
- Some remain as stem cells; others become primary spermatocytes
- Ensures continuous sperm production throughout life

Spermatogenesis

- **Primary spermatocytes** (初级精母细胞) undergo meiosis I
- Form **secondary spermatocytes** (次级精母细胞)
- Secondary spermatocytes undergo meiosis II
- Form haploid **spermatids** (精子细胞)

Spermatogenesis

- Spermatogenesis occurs in stages within germinal epithelium
- Sertoli cells provide support, nutrients, and regulation
- **Blood-testis barrier** (血睾屏障) protects developing sperm
- Tight junctions between Sertoli cells create immune privilege

Spermatogenesis

- **Spermiogenesis** (精子形成) transforms spermatids into spermatozoa
- Acrosome formation over nucleus
- Nuclear condensation
- Mitochondrial arrangement in midpiece
- Tail development
- Excess cytoplasm discarded as residual bodies

Spermatogenesis

- Mature spermatozoon has three regions
- **Head**: nucleus with condensed DNA; **acrosome** (顶体) with enzymes
- **Midpiece**: mitochondria providing ATP
- **Tail**: flagellum for motility
- Highly specialized for swimming to and fertilizing ovum

Male Accessory Organs

- Male reproductive system includes testes, ducts, and glands
- **Epididymis** (附睾): sperm maturation and storage
- **Ductus deferens** (输精管): transports sperm
- **Seminal vesicles** (精囊): fructose, prostaglandins (60% semen)
- **Prostate** (前列腺): alkaline fluid (30% semen)
- **Bulbourethral glands** (尿道球腺): mucus for lubrication

Male Accessory Organs

- **Penis** (阴茎) contains erectile tissue
- Two **corpora cavernosa** (阴茎海绵体)
- One **corpus spongiosum** (尿道海绵体)
- Erectile tissue has vascular spaces that fill with blood

Erection Mechanism

- **Erection** (勃起) from parasympathetic stimulation
- **Nitric oxide (NO)** (一氧化氮) is key mediator
- NO activates guanylate cyclase → increases cGMP
- cGMP closes $\mathrm{Ca}\hat{}\{2+\}$ channels → smooth muscle relaxation
- Vasodilation allows blood to fill erectile tissue

Erection Mechanism

- **PDE5** (phosphodiesterase type 5, 5 型磷酸二酯酶) breaks down cGMP
- **Sildenafil (Viagra)** (西地那非) inhibits PDE5
- Prolongs cGMP action and enhances erection
- Treats **erectile dysfunction** (勃起功能障碍)

Erection Mechanism

- **Emission** (射出): movement of sperm/secretions to urethra (sympathetic)
- **Ejaculation** (射精): expulsion of semen through contractions
- **Semen** (精液): 100-400 million sperm per ejaculate
- Contains sperm plus glandular secretions

Male Fertility

- Normal semen parameters important for fertility
- Volume: 2-5 mL
- Sperm count: >20 million/mL
- Motility: >50% motile
- Morphology: >30% normal shape

Male Fertility

- **Vasectomy** (输精管切除术) is surgical sterilization
- Segment of ductus deferens removed or blocked
- Prevents sperm from entering semen
- Does not affect testosterone or sexual function

Outline

Female Reproductive Anatomy

- **Ovaries** (卵巢): produce ova and hormones
- **Uterine tubes** (输卵管): transport ova; site of fertilization
- **Uterus** (子宫): environment for embryo and fetus
- **Cervix** (子宫颈): neck of uterus
- **Vagina** (阴道): canal to external environment

Female Reproductive Anatomy

- Sagittal view shows anatomical relationships
- Reproductive organs positioned between bladder and rectum
- Uterus normally anteverted (tilted forward)
- Understanding anatomy essential for clinical practice

Female Reproductive Anatomy

- External genitalia (**vulva**, 外阴) includes:
- **Labia majora** (大阴唇): homologous to scrotum
- **Labia minora** (小阴唇): smaller inner folds
- **Clitoris** (阴蒂): homologous to penis
- Demonstrates homology between male and female structures

Ovarian Follicle Development

- Ovaries contain follicles at various developmental stages
- **Primordial follicles** (原始卵泡): primary oocyte arrested in prophase I
- **Primary follicles** (初级卵泡): oocyte with granulosa cells
- **Secondary follicles** (次级卵泡): fluid-filled spaces develop
- **Graafian follicle** (格拉夫卵泡): mature follicle with large antrum

Ovarian Follicle Development

- Primary oocyte resumes meiosis just before ovulation
- Completes meiosis I → secondary oocyte + first polar body
- Secondary oocyte arrested in metaphase II
- Meiosis II completed only if fertilization occurs

Ovarian Follicle Development

- Ovary contains follicles at different stages simultaneously
- Growing follicles, mature follicle, corpus luteum
- **Atretic follicles** (闭锁卵泡): dying by apoptosis
- **Corpus albicans** (白体): scar tissue from degenerated corpus luteum

Ovulation

- **Ovulation** (排卵) is release of secondary oocyte
- Triggered by LH surge at mid-cycle
- Oocyte released with surrounding corona radiata
- Cloud of follicular fluid accompanies release

Ovulation

- **Oogenesis** (卵子发生) differs from spermatogenesis
- Begins during fetal development
- Primary oocytes arrested until puberty
- Monthly resumption: meiosis I → secondary oocyte + first polar body
- Meiosis II completed at fertilization → ovum + second polar body

Ovulation

- Monthly ovarian cycle progresses through stages
- Primordial → primary → secondary → graafian follicle
- Ovulation releases oocyte
- Corpus luteum forms from ruptured follicle
- Degenerates to corpus albicans if no pregnancy

Hormonal Regulation

- FSH stimulates follicle growth and estrogen secretion
- LH stimulates theca cells to produce androgens
- Granulosa cells convert androgens to estrogen
- Rising estrogen initially inhibits FSH/LH (negative feedback)
- High estrogen triggers LH surge (positive feedback) → ovulation

Outline

Hormonal Changes

- **Menstrual cycle** (月经周期) averages 28 days
- Coordinated changes in FSH, LH, estrogen, progesterone
- LH surge at mid-cycle triggers ovulation
- Hormonal patterns ensure endometrial preparation for implantation

Hormonal Changes

- Integration of pituitary, ovarian, and endometrial cycles
- Pituitary hormones drive ovarian changes
- Ovarian hormones regulate endometrial changes
- Downward arrows show hormonal effects

Ovarian Cycle Phases

- **Follicular phase** (卵泡期): Days 1-14
  - FSH stimulates follicle development
  - Rising estrogen from growing follicles
  - Ends with LH surge and ovulation
- **Luteal phase** (黄体期): Days 14-28
  - Corpus luteum secretes progesterone and estrogen
  - If no pregnancy, corpus luteum degenerates
  - Hormone drop triggers menstruation

Ovarian Cycle Phases

- **Corpus luteum** (黄体) is yellowish endocrine structure
- Forms from ruptured follicle after ovulation
- Secretes progesterone and estrogen
- Essential for maintaining early pregnancy

Ovarian Cycle Phases

- Endocrine control involves feedback loops
- Negative feedback dominates most of cycle
- Positive feedback triggers LH surge at mid-cycle
- Integration of hypothalamic, pituitary, and ovarian signals

Ovarian Cycle Phases

- Menstrual cycle phases clearly defined
- Each phase characterized by specific hormonal patterns
- Endometrial changes correspond to ovarian events
- Understanding phases essential for fertility management

Endometrial Changes

- **Endometrium** (子宫内膜) undergoes cyclic changes
- **Menstrual phase** (月经期): Days 1-5, shedding of functional layer
- **Proliferative phase** (增殖期): Days 5-14, estrogen-stimulated growth
- **Secretory phase** (分泌期): Days 14-28, progesterone-stimulated secretion
- Without implantation, progesterone withdrawal causes menstruation

Temperature and Fertility

- **Basal body temperature** (基础体温) rises after ovulation
- Increase of 0.3-0.5°C due to progesterone
- Can be used for fertility awareness
- Helps identify ovulation timing for natural family planning

Factors Affecting Cycles

- **Pheromones** (信息素) may synchronize cycles among women
- Stress disrupts cycles through effects on GnRH
- Body fat affects reproduction:
  - Very low fat ($<$17%) can cause **amenorrhea** (闭经)
  - Obesity disrupts cycles through hormone metabolism
- Minimum body fat required for normal cycles

Contraception and Menopause

- Contraceptive methods include:
  - Oral contraceptives: suppress ovulation
  - Barrier methods: block sperm
  - IUD: prevents implantation
  - Sterilization: tubal ligation or vasectomy
- Choice depends on effectiveness, side effects, and preferences

Contraception and Menopause

- **Menopause** (更年期) typically age 45-55
- Cessation of cycles for 12 consecutive months
- Ovarian follicles depleted; estrogen/progesterone decrease
- FSH and LH rise due to loss of negative feedback
- Symptoms: hot flashes, vaginal dryness, mood changes, bone loss
- **HRT** (hormone replacement therapy, 激素替代疗法) can help but has risks

Outline

Fertilization Process

- Fertilization occurs in ampulla of uterine tube
- **Capacitation** (获能) of sperm in female tract (6-8 hours)
- Sperm must penetrate corona radiata and zona pellucida
- Only capacitated sperm can undergo acrosome reaction

Fertilization Process

- Fertilization sequence:
  - Sperm encounters corona radiata
  - Acrosome reaction releases enzymes
  - Penetration through zona pellucida
  - Fusion with oocyte membrane

Fertilization Process

- **Acrosome reaction** (顶体反应) essential for fertilization
- Acrosome contains **hyaluronidase** (透明质酸酶) and **acrosin** (顶体酶)
- Enzymes digest path through protective layers
- Acrosomal membrane fuses with plasma membrane

Fertilization Process

- Upon fertilization, secondary oocyte completes meiosis II
- Forms second polar body
- Male and female pronuclei fuse
- **Cortical reaction** (皮质反应) prevents polyspermy
- Zona pellucida modified to block additional sperm

Fertilization Process

- **In vitro fertilization (IVF)** (体外受精) occurs outside body
- **ICSI** (intracytoplasmic sperm injection, 卵胞浆内单精子注射)
- Single sperm injected directly into oocyte
- Used for severe male infertility

Early Development

- **Cleavage** (卵裂) begins immediately after fertilization
- Rapid mitotic divisions without cell growth
- Zygote $\rightarrow$ 2 $\rightarrow$ 4 $\rightarrow$ 8 cells (day 3)
- **Morula** (桑椹胚): 16-32 cells (day 3-4)
- **Blastocyst** (囊胚): hollow ball (day 5-6)

Early Development

- Scanning electron micrographs show early stages
- 4-cell stage shows individual blastomeres
- 16-cell morula stage shows compacted cells
- These stages occur during transport through uterine tube

Early Development

- Blastocyst structure:
  - **Trophoblast** (滋养层): outer layer, forms placenta
  - **Inner cell mass** (内细胞团): becomes embryo proper
  - **Blastocoel** (囊胚腔): fluid-filled cavity
- Differentiation begins before implantation

Implantation

- **Implantation** (着床) begins day 5-7, completed by day 10
- Blastocyst adheres to endometrium
- Trophoblast invades uterine tissue
- **Syncytiotrophoblast** (合体滋养层) invades endometrium
- **Cytotrophoblast** (细胞滋养层) remains as cellular layer

Implantation

- **hCG** (human chorionic gonadotropin, 人绒毛膜促性腺激素) from trophoblast
- Maintains corpus luteum during first trimester
- Ensures continued progesterone secretion
- Detected in blood/urine - basis for pregnancy tests
- Peaks at 8-10 weeks

Placenta Formation

- Placenta develops from trophoblast and maternal tissue
- **Chorionic villi** (绒毛) are fingerlike projections
- Contain fetal blood vessels
- Increase surface area for maternal-fetal exchange

Placenta Formation

- Placental structure:
  - **Amnion** (羊膜) surrounds fetus in amniotic fluid
  - **Chorion frondosum** (绒毛膜): fetal portion
  - **Decidua basalis** (蜕膜底层): maternal portion
  - **Umbilical cord** (脐带): two arteries, one vein

Placenta Formation

- **Amniocentesis** (羊膜穿刺术) detects genetic disorders
- Performed at 15-18 weeks
- Amniotic fluid withdrawn for analysis
- Chromosomal and biochemical testing possible

Placenta Formation

- Blood circulation within placenta
- Maternal and fetal blood separated but do not mix
- Exchange occurs across placental barrier
- Fetal blood: umbilical arteries (deoxygenated) → villi → umbilical vein (oxygenated)

Placenta Formation

- Exchange across placenta by multiple mechanisms:
  - Diffusion: O_2, CO_2, nutrients, wastes
  - Active transport: amino acids, vitamins, minerals
  - Receptor-mediated endocytosis: IgG antibodies
- Placenta barrier protects but not absolute
- Alcohol, drugs, some viruses can cross

Placental Endocrine Function

- Placenta produces hormones requiring fetal-maternal cooperation
- Maternal cholesterol → placental progesterone
- Fetal DHEAS → placental estrogens
- Demonstrates integrated fetal-placental-maternal unit

Placental Endocrine Function

- Major placental hormones:
  - hCG: maintains corpus luteum
  - hCS: promotes growth, metabolic changes
  - Estrogens: uterine growth, breast development
  - Progesterone: maintains endometrium, inhibits contractions

Labor and Parturition

- **Parturition** (分娩) is childbirth process
- **Labor** (产程) involves uterine contractions
- Positive feedback mechanism initiates labor
- Fetal adrenal → DHEAS and cortisol
- Cortisol → placental CRH → more cortisol (positive feedback)
- Estriol, prostaglandins, oxytocin → contractions

Labor and Parturition

- Three stages of labor:
  - Stage 1: cervical dilation (8-24 hours)
  - Stage 2: delivery of baby (30 min - 2 hours)
  - Stage 3: delivery of placenta (5-30 minutes)
- **Oxytocin** (催产素) stimulates contractions
- Contractions → push fetus → stretch cervix → more oxytocin (positive feedback)

Lactation

- Mammary gland structure:
  - **Lobules** (小叶) with **alveoli** (腺泡) produce milk
  - **Lactiferous ducts** (输乳管) transport milk to nipple
- Development during pregnancy prepares for lactation

Lactation

- Hormonal control of lactation:
  - During pregnancy: estrogen/ progesterone stimulate development but inhibit milk production
  - After delivery: hormone drop allows prolactin to stimulate milk production
  - High estrogen during pregnancy blocks prolactin's lactogenic effect

Lactation

- Lactation involves two processes:
  - **Milk production**: stimulated by prolactin from anterior pituitary
  - **Milk ejection**: stimulated by oxytocin from posterior pituitary
- Suckling triggers neuroendocrine reflex releasing both hormones

Lactation

- Maternal antibodies protect newborn:
  - IgG crosses placenta during pregnancy (protects 3-12 months)
  - IgA in colostrum and breast milk protects intestine
  - **Colostrum** (初乳): first milk, rich in antibodies
- Passive immunity bridges gap until infant immune system matures

Outline

Summary

- Sexual reproduction maintains genetic diversity through meiosis and fertilization
- Sex determination by SRY gene; hormones direct sexual differentiation
- HPG axis regulates reproduction through feedback loops
- Male reproduction: continuous spermatogenesis; testosterone production
- Female reproduction: cyclic ovulation; coordinated hormonal changes
- Fertilization initiates development; implantation establishes pregnancy
- Placenta mediates maternal-fetal exchange and hormone production
- Parturition initiated by positive feedback; lactation nourishes newborn